

Global Survey on Current Practices of Endobronchial Ultrasound Approaches to N2 Staging in Non-Small Cell Lung Cancer Following the Ninth Edition of the TNM Classification

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Background: The ninth edition of the TNM classification (TNM-9) for NSCLC introduces distinct N2 subgroups based on single (N2a) or multiple (N2b) involved stations, impacting prognosis and staging. This survey aimed to assess the impact of these changes on mediastinal staging with EBUS-TBNA, particularly regarding the need for needle change when sampling lymph nodes from different N2 stations.

Methods: A 10-question online questionnaire was distributed from March 4, 2025 to March 16, 2025, to pulmonologists through interventional pulmonology sections of medical societies, online networks, and social media.

Results: A total of 605 questionnaires from 66 countries were analyzed. Most respondents were males (58.8%), aged 40 to 49 years, and worked in academic hospitals. Regarding EBUS procedure

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Ethics approval was not deemed necessary due to the anonymous and voluntary nature of the surveys.

A separate text was provided prior to the survey, informing participants that the results would be published and that participation was voluntary. Therefore, participation was considered as consent for publication.

K.K., A.B., R.E., and F.C.T.: conceptualized the study and developed the questionnaire. F.C.T., K.K., and R.E.: analyzed and grouped the free-text comments. P.H.: assisted in creating the graphics. F.C.T.: performed the statistical analysis. F.C.T. and K.K.: wrote the first draft of the manuscript. K.K., A.B., R.E., F.C.T., L.C.M., K.B., E.H.F.M.v.d.H., and P.L.: distributed the questionnaire in their countries and professional societies. All authors reviewed, revised the manuscript, and agreed to the final version.

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frequency, 28.9% performed 1 to 2 procedures weekly, and 13.6% performed more than 10. When asked about the need for needle change between different N2 stations, 20.7% saw it as necessary, 33.2% disagreed, and 17.4% considered it necessary in some cases. Although 38.1% considered changing the needle at least sometimes, only 5.5% reported actually doing so. The main barriers were costs (68.3%) and lack of evidence (31.3%). Those against needle change cited the belief that it does not affect results (58.2%), lack of evidence (54.2%), and other factors (4.5%), with many considering flushing the needle sufficient. Free-text responses indicated a lack of knowledge or implementation of TNM-9 in some centers.

Conclusion: Opinions on implementing TNM-9 are inconsistent, within the IP community, particularly regarding needle change for different N2 stations. Education, reliable data, and consensus guidelines are needed.

Key Words: non-small cell lung cancer, NSCLC, EBUS, ultrasound-guided transbronchial needle aspiration, EBUS-TBNA, TNM classification

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The ninth edition of the TNM classification, which has been formally adopted since January 2025, introduces important changes and refinements regarding the lymph node (N) status.¹ The N component in lung cancer has historically been classified based on the anatomic location of the lymph nodes, without considering the number of lymph node stations involved. Quantifying nodal disease burden has been recognized, in exploratory analyses performed for the seventh and eighth TNM editions, as a crucial factor affecting the disease prognosis. The ninth edition of the TNM staging classification was based on a database of 124,581 cases, with 87,043 cases being enrolled in the analysis, with cases compiled from 75 different sites across 25 countries. Of patients, 56% came from Asia and Australia, 27.7% from Europe, and 15.7% from North America.^{2,3} The analysis identified clearly distinct and consistent prognostic discrimination of subgroups in patients with N2 disease, specifically between single-station involvement (N2a) and multiple-station involvement (N2b). These observations lead to the subdivision of the N2 mediastinal lymph nodes into N2a and N2b.⁴

This new N2 subdivision may result in a refinement of current standards and practices in mediastinal staging using EBUS-TBNA in interventional pneumology. It is now of great importance to clearly distinguish between the subcategories of N2 lymph nodes, because an accurate mediastinal staging, with differentiation between N2a and N2b status, has significant therapy and also prognostic implications. It directly affects the tumor stage—whether it is IIB, IIIA, or IIIB. The new classification, based on data from the International Association for the Study of Lung Cancer (IASLC), clearly highlights the prognostic significance of N2 subcategories. Survival data show that in non-small cell lung cancer (NSCLC), single-site N2 involvement is associated with a better prognosis (N2a overall survival 47%) compared with multisite N2 involvement (N2b overall survival 36%).² Furthermore, for patients who are candidates for definitive chemoradiotherapy, this N2 subdivision also directly influences the radiation treatment field.

Therefore, for those performing mediastinal staging, N2 is no longer a straightforward classification but must be differentiated between two distinct N2 lymph node stations. This change raises several questions regarding the practical

procedure. First, it is clear that a systematic evaluation of the mediastinal and hilar lymph nodes is indicated—versus the so-called “hit and run approach.” Subsequently, which of the N2 lymph nodes should be punctured first? Is the decision based solely on anatomic and size considerations (eg, ≥ 5 mm)? Should it be guided by suspected B mode criteria and/or by the positron emission tomography-computed tomography (PET-CT) enhancement of the lymph nodes? However, PET avidity depends on size and tumor entity and EBUS-TBNA is essential for confirming PET negative mediastinal metastases, especially in adenocarcinoma.^{5,6} Furthermore, the probability of lymph node involvement depends on the size and localization of the primary tumor.⁷ Given the lack of standardized guidelines for mediastinal staging based on the ninth edition of the TNM staging classification and the numerous practical issues arising from this, we conducted a worldwide survey of practicing physicians who perform EBUS, to assess current practice in interventional pulmonology and to gather expert opinion and insight on this important topic, and to discuss whether changing the EBUS needle between different N2-lymph node stations is indicated to avoid incorrect upstaging and/or tumor cell spread and consequent upstaging.

METHODS

Study Design and Participants

An online questionnaire was developed using the LimeSurvey platform, consisting of 10 questions about the age groups of the participants, the institutions/clinics and country where the participants work and the current approach for N2 staging. The questionnaire also explores individual interventionist's opinion of how N2 staging should be performed in light of ninth TNM staging, the barriers to performing a needle exchange, and the availability of PET-CT before the procedure. The development of the questionnaire was carried out by the core group (including K.K., F.C.T., R.E., and A.B.). Each question also provided an opportunity for respondents to share their opinions in free-text fields. All questionnaires to which the participants gave at least one response were included in the final analysis; completely blank questionnaires were excluded from the analysis. The complete questionnaire can be found in the appendix (Supplementary File S1, Supplemental Digital Content 1, <http://links.lww.com/LBR/A365>).

Distribution of the Questionnaire through International and National Networks

The questionnaire was distributed to interventional pulmonologists between March 4 and 16, 2025, through direct outreach and sharing within professional networks of interventional pulmonologists. The questionnaire was distributed through various online networks of interventional pulmonology groups and through email through international professional societies, including the European Respiratory Society (ERS) Assembly 14 “Clinical techniques, imaging and endoscopy,” and the European Association for Bronchology and Interventional Pulmonology (EABIP). In addition, it was shared through distribution lists of national pulmonology and bronchology associations, as well as through social media platforms such as LinkedIn and Facebook.

Qualitative Analysis of Free-text Responses

Alongside the categorical questions on whether changing the needle during EBUS staging between two N2 lymph nodes when suspecting an N2 stage was considered necessary (“yes,” “no,” “sometimes”), participants had the option to provide a comment explaining their choice.

The collected comments were categorized based on similar content by 2 pulmonologists (K.K. and F.C.T.). In cases of disagreement between the two, a third pulmonologist (R.E.) was consulted to make a final decision. This approach allowed for a systematic capture of the participants’ different perspectives and reasons, grouping the qualitative responses into coherent categories. The categorization enabled the identification of patterns and key points in the comments and provided a foundation for analyzing the participants’ opinions.

Statistical Analysis

Data in the tables are presented as counts and percentages or as means and SDs. All analyses were conducted using SPSS version 25 (IBM Corp.), with $P < 0.05$ considered statistically significant.

RESULTS

Survey Participants

A total of 896 participants from 66 different countries took part, of which only 605 questionnaires could be included in the analysis due to missing values. The majority of participants, 354 (58.5%), were males, 177 (29.3%) were females, and 74 (12.2%) did not specify their sex. Three hundred thirty-four participants (55.2%) came from academic/university hospitals, 81 (13.4%) from general hospitals, 48 (7.9%) from specialist clinics, and 7 (1.2%) from other settings. One hundred thirty-five (22.3%) did not answer this question. The majority of respondents were members of one or more scientific societies. Four hundred sixty-three (73.2%) of those who answered the question were members of the ERS, 162 (26.8%) were members of the EABIP, 188 (31.1%) of the World Association for Bronchology and Interventional Pulmonology (WABIP), 76 (12.6%) of the American Thoracic Society (ATS), and 184 (30.4%) were members of other medical societies. The majority of participants reported performing mediastinal staging using EBUS-TBNA regularly: 175 (28.9%) performed 1 to 2 procedures per week, 169 (27.9%) performed 3 to 5 procedures per week, 111 (18.3%) performed 6 to 10 procedures per week, and 82 (13.6%) performed more than 10 procedures per week.

Figure 1 shows the geographic origin of the respondents who provided information about their background. The characteristics of the respondents are listed in Table 1.

Current Approach, Thoughts, and Challenges Regarding the Practical Implementation in the Evaluation of N2 Lymph Node

Regarding the question, “When puncturing 2 lymph nodes in the N2 category, do you change the needle between the punctures?”, 164 respondents (27.1%) did not answer, only 33 (5.5%) answered “yes,” 347 (57.4%) answered “no,” and 61 (10.1%) answered “sometimes”. In contrast, the general question, “Do you believe it is important to change the needle between the punctures of two N2 lymph nodes?” was answered by 125 respondents (20.7%) with “yes,” 201 respondents (33.2%) with “no,” and 105 respondents

(17.4%) with “sometimes.” One hundred seventy-four respondents (28.8%) did not answer. Of the 230 respondents who considered a needle change to be at least sometimes advisable, 157 (68.3%) indicated that they do not perform it due to cost reasons, while 72 (31.3%) stated that there is a lack of evidence for such an approach. In addition, 17 respondents (7.4%) reported “other reasons” (Table 2 and Fig. 2).

Availability of PET-CT Before Bronchoscopy in the Centers

Regarding the question, “Is a PET-CT scan a routine examination before the first bronchoscopy for patients with suspected tumors in your hospital?” 71 respondents (11.7%) answered “always,” 239 respondents (39.5%) answered “most of the time,” 120 respondents (19.8%) answered “rarely,” and 23 respondents (3.8%) answered “never.” In addition, 152 respondents (25.1%) did not answer the question (Table 2).

Analysis of Free-text Responses

In addition to the response options in the various categories, participants had the opportunity to provide free-text comments. All free-text comments can be found in the Supplemental Tables S2 to S4 (Supplemental Digital Content 1, <http://links.lww.com/LBR/A365>). Those who generally opposed a needle change expressed their views in 5 main areas. (1) View that a needle change is not necessary when following a systematic approach: “*Sample least to most likely involved. Needle contamination is overstated*” (Participant 225). (2) The view that flushing or disinfecting the needle is sufficient: “*Washing is enough.*” (Participant 170) “*If the needle is properly flushed, the contamination possibility is almost none*” (Participant 360). (3) Lack of knowledge or acceptance of the TNM ninth edition: “*It is not necessary to change the needle as all N2s are in the same stage*” (Participant 72). (4) Lack of evidence “*Because the needle is cleaned from the sample after every puncture and there are no data to support that even 1 cancer cell can be transferred or even grow in population inside a lymph node, there are no experiments even in mice up to date*” (Participant 661). (5) Local standards, costs, and ecological aspects: “*No need to.*” (Participant 694) “*...Needle contamination is overstated*” (Participant 225). “*Changing the needle is not cost-efficient* (Participant 469).” “*if so much contamination between puncturing different lymph nodes would remain in the needle, we would have to use for each lymph node a separate needle. That is economically and ecologically not useful*” (Participant 474). “*Have not been advised to do so*” (Participant 702). For those who considered changing the needle necessary, the main argument was based on (1) the rationale of the TNM ninth edition and the risk of upstaging. “*There is need to change the needle to avoid staging false multiple N2*” (Participant 20). “*It defines the N2 stage appropriately*” (Participant 87). However, participants in this group also mentioned concerns regarding the costs: “*Solo que en mi país depende de la parte económica*” (Translation: “*Only that in my country it depends on the economic side.*” *Changing the needle increases costs unnecessarily*). One participant mentioned the lack of expert consensus: “*I believe it could be important, but when I or somebody else asked about it at conferences I attended, there was no clear statement about it.*” Some centers do it (or say they do), others don’t” (Participant 653). Those who answered “sometimes” regarding the needle change

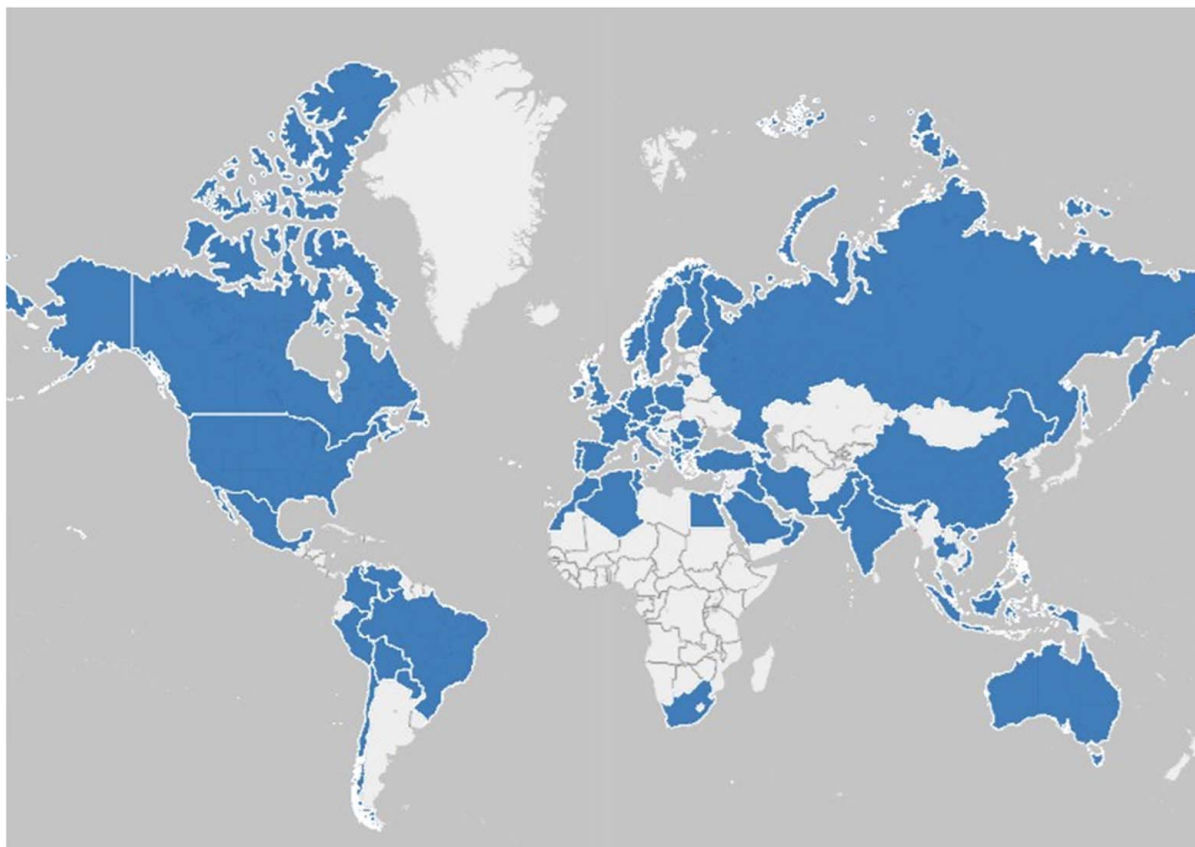


FIGURE 1. Map of the countries of origin of survey participants. This map displays the geographical distribution of the countries from which the survey participants originated.

reasoned their decision with the following: (1) Decision based on lymph node morphology (ultrasound, CT, PET): “*If concern on PET regarding multiple N2*” (Participant 535). “*When there are suspicious ultrasound features*” (Participant 530). (2) Decision based on ROSE results and availability: “*positive ROSE*” (Participant 374). “*Yes, if one lymph node is with found with malignancy on ROSE and an additional lymph node afterwards also has to be sampled*” (Participant 575). (3) Decision based on suspected tumor stage and therapeutic consequences: “*Differentiate between IIB and IIIA*” (Participant 664). “*Depends on staging and treatment goals*” (Participant 777). Participants in this group also stated the need for more research, along with concerns about the high costs involved: “*Scientific evidence is too scarce*” (Participant 16). “*Should be studied*” (Participant 170). “*There might be cases where it is useful, but the needles are too expensive to do it on a regular basis*” (Participant 465).

DISCUSSION

The aim of the present survey was to evaluate current practice and gather insights from physicians performing EBUS and other interventional pulmonology procedures, acknowledging the lack of standardized guidelines for endoscopic mediastinal staging following the ninth edition of the TNM classification. The survey was conducted among a large number of interventional pulmonologists worldwide who perform mediastinal lung cancer staging

using EBUS-TBNA. Most respondents reported performing EBUS staging on a regular basis, the majority were over 40 years of age, members of one or more medical societies, and employed at a university institution, so we assume that they are experienced, dedicated, and scientifically interested bronchoscopists.

The main findings of the study were: Interventional pulmonology is divided on the practical implementation of the new staging classification in the case of a possible N2 stage. The number of those who rejected a needle change in general was about the same as the number of those who considered a needle change useful either in general or at least in individual cases. Another finding is that the new classification has not yet been embraced by all colleagues, and not everyone was aware of the new N2 subclassification or that its introduction would mean challenging the traditional systematic approach to EBUS staging. In addition, the majority of respondents, including those who stated that, in theory, they consider a needle exchange to be beneficial either in general or in specific cases, reported nevertheless that they would not implement it in practice. The main arguments in favor of needle exchange were the changes associated with the ninth edition of TNM and the fear of tumor cell dissemination, potentially leading to iatrogenic upstaging.

Arguments against the practice included the lack of scientific evidence to support it, the assumption that the current systematic approach was sufficient, and the belief that rinsing the needle with normal saline was adequate. The

TABLE 1. Characteristics of the Study Participants, N = 605

Variable	N (%)
Sex	
Female	177 (29.3)
Male	354 (58.5)
No Answer	74 (12.2)
Age groups (y)	
Under 30	12 (6.9)
30-39	139 (23.0)
40-49	229 (37.9)
50-59	130 (21.5)
Over 60	53 (8.8)
No answer	42 (6.9)
Institutional affiliation of respondents	
Academic/university hospital	334 (55.2)
General hospital	81 (13.4)
Specialized clinic	48 (7.9)
Other (please specify)	7 (1.2)
No answer	135 (22.3)
No. TBNA procedures per week	
< 1 or no answer	68 (11.2)
1-2	175 (28.9)
3-5	169 (27.9)
6-10	111 (18.3)
More than 10	82 (13.6)
Membership in scientific societies*	
ERS	463 (73.2)
EABIP	162 (26.8)
WABIP	188 (31.1)
ATS	76 (12.6)
Others	184 (30.4)

*Multiple answers allowed.

available data on this topic remain limited, but there are indications of needle contamination in the literature. One published study by Puyal et al prospectively analyzed 62 patients undergoing linear endobronchial ultrasound for suspected lung tumors. Despite multiple needle flushes (up to 10 mL saline), cytologic analysis revealed persistent tumor cell contamination in 43.5% of cases, with no significant differences related to patient or tumor.⁸ A second study, presented as a poster at the 2017 ATS Congress, also confirmed the presence of needle contamination. In that study, flushing the needle with 10 mL saline did not eliminate contamination, leading the authors to recommend sequential staging to mitigate this issue.⁹

In the free-text comments, it was also noticeable that several colleagues, such as participant 72, who stated, “*It is not necessary to change the needle as all N2s are in the same stage,*” were either unaware of the new TNM classification or had not incorporated it into their endoscopy practices.

Among those who considered a needle exchange to be beneficial in theory, the main reason they did not implement it in practice was the associated high costs. Time savings, in contrast, were considered less important. In addition, even among colleagues who deemed the needle exchange to be sensible, there was a desire for more evidence to justify such an approach. The numerous free-text comments, particularly within the group of those who considered a needle exchange beneficial in certain cases, show that many respondents have critically engaged with the topic.

We observe a remarkable difference in clinical practice: where in the clinic the subject of cross-contamination between multiple N2 stations is not standardized and

debated as described previously, in pathology laboratories processing our cytologic specimens, are required to implement strict protocols preventing cross-contamination as following the Clinical Laboratory Improvement Amendments, and the Joint Commission and the College of American Pathologists.¹⁰ Moreover, while discussing using multiple needles for different N2 stations, in some regions of the world, sometimes the single-use designed needles are even resterilized and reused despite recommendations against such practices.¹¹

In addition, it can be assumed that PET-CT before bronchoscopy and ROSE could gain even more importance in the future. PET-CT plays an important role in helping to select nodal biopsy targets to confirm the highest possible disease stage and can help with the selection of target nodes for the EBUS-TBNA.¹² However, the imaging modalities have significant limitations. Specifically, false positive results may arise due to inflammation. In addition, nodal metastases that are not FDG-avid because of central necrosis or tumor size or tumor biology, present another potential pitfall.⁵ According to the American Society of Clinical Oncology (ASCO) recommendations, the inherent limitations in the accuracy of imaging alone require that lymph nodes, including those that are Fluorodeoxyglucose (FDG)-avid, which raise suspicion for metastases, must undergo pathologic confirmation when possible, especially when curative treatment is intended.^{11,12} In light of induction treatments based on chemotherapy and immunotherapy for potentially resectable lung cancers, the accurate discrimination of N2a and N2b involvement between N2a and N2b disease will help better select patients who would maximally benefit from surgical resection. Some colleagues suggested a pragmatic approach, ROSE to differentiate between different N2 lymph nodes. They recommended

TABLE 2. Current Approach, Thoughts, and Challenges Regarding the Practical Implementation in the Evaluation of N2 lymph node

Variable	N (%)
When puncturing 2 lymph nodes in the N2 category, do you change the needle between the punctures?	
No answer	164/606 (27.1)
Yes	33/605 (5.5)
No	347/605 (57.4)
Sometimes	61/605 (10.1)
Do you believe it is important to change the needle between the punctures of two N2 lymph nodes?	
No answer	174/605 (28.8)
Yes	125/605 (20.7)
No	201/605 (33.2)
Sometimes	105/605 (17.4)
Reasons for not changing the needle among those who consider it at least sometimes advisable (N = 230)	
Costs	157/230 (68.3)
Lack of evidence for such an approach	72/230 (31.3)
Other reasons	17/230 (7.4)
Is a PET-CT scan a routine examination before the first bronchoscopy for patients with suspected tumors in your hospital?	
No answer	152/605 (25.1)
Always	71/605 (11.7)
Most of the time	239/605 (39.5)
Rarely	120/605 (19.8)
Never	23/605 (3.8)

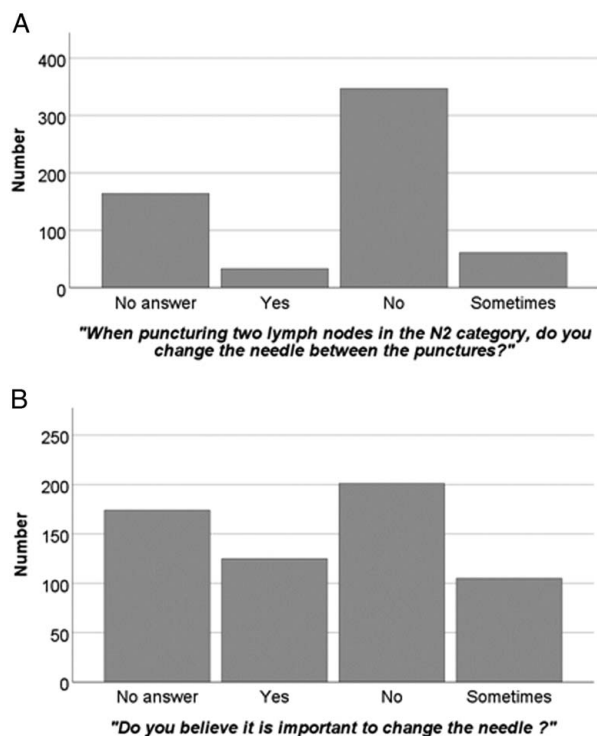


FIGURE 2. EBUS needle management during N2 lymph node puncture: perspectives and practical approach. The diagrams provide an overview of the decision-making process regarding needle management during N2 lymph node punctures. A, The survey responses on whether the needle should be changed between puncturing two lymph nodes. B, Outlines the actual practices reported from clinical settings.

changing the needle only when suspicious findings are identified during the ROSE procedure. However, this approach requires the availability of ROSE at the site, which was not explicitly addressed in the survey. Based on our understanding, we assume that the majority of sites do not have this option. In contrast, the availability of PET-CT was queried, and the results show that the majority of respondents indicated that at least in certain cases, PET-CT is already available at the time of the bronchoscopy. However, in some health care systems, PET-CT can only be performed after histologic confirmation. Nonetheless, there are still many open questions, and the general consensus in interventional pulmonology is that there is a lack of valid data to provide clear guidelines regarding the needle exchange, which would be associated with significant costs.

Given the lack of definitive clinical evidence, our findings should be interpreted as highlighting variability in current practice and the need for well-designed prospective studies, including cost-effectiveness analyses, to determine whether changing needles between lymph node stations is warranted. A meaningful study on this topic would need to go beyond the purely cytologic demonstration of tumor cell carryover and, in light of the increasing use of neoadjuvant therapies, demonstrate a true clinical benefit of a cost-intensive approach such as needle exchange between lymph node stations. The current inconsistency in standards and the uncertainty surrounding the approach highlight the need for such evidence.

Limitations

A major limitation of our study is the inability to determine the total number of individuals who received the survey and the corresponding response rate, due to the informal nature of its distribution through professional networks. In addition, because the survey was anonymous and distributed online through multiple channels, the possibility of duplicate responses cannot be fully excluded. Furthermore, the survey was specifically targeted at interventional pulmonologists and, therefore, not distributed through thoracic surgical societies or the American College of Chest Physicians (ACCP). This may have introduced selection bias and limits the generalizability of our findings.

In addition, a large number of participants either did not answer all of the questions or had to be excluded from the analysis due to not answering any of the questions. For data privacy reasons, we collected only minimal information about the respondents. The questionnaire was intentionally kept very brief; however, it would certainly have been beneficial to gather more detailed information regarding the setting and local standards for the evaluation of lung cancer. Despite these limitations, the survey was successful in reaching a large number of experienced interventional pulmonologists and provides valuable insights into the uncertainties, challenges, and practices in endoscopy worldwide, particularly regarding the implementation of the N2 subclassification.

CONCLUSION

The survey aimed to assess current practices in interventional pulmonology for mediastinal staging using EBUS-TBNA, in light of the TNM ninth Edition N2 subclassification. Conducted globally, it revealed significant uncertainties and disagreements regarding the practice of changing the EBUS needle for different N2 lymph nodes, with the majority of respondents not implementing this in practice. While about half of the respondents considered it sensible in theory, at least in selected cases, financial constraints were the primary reason for not changing the needle. Free-text responses suggested using PET-CT or ROSE to guide needle changes, although the availability of both is limited, and clear procedural guidelines from professional societies are lacking. Further research is needed, focusing on the risk of cross-contamination in sequential sampling of multiple (N3 and N2) lymph nodes, and potential role of ROSE therein. A joint expert statement from ERS, EABIP, WABIP, EACTS, and ATS would definitely be beneficial to standardize the approach.

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